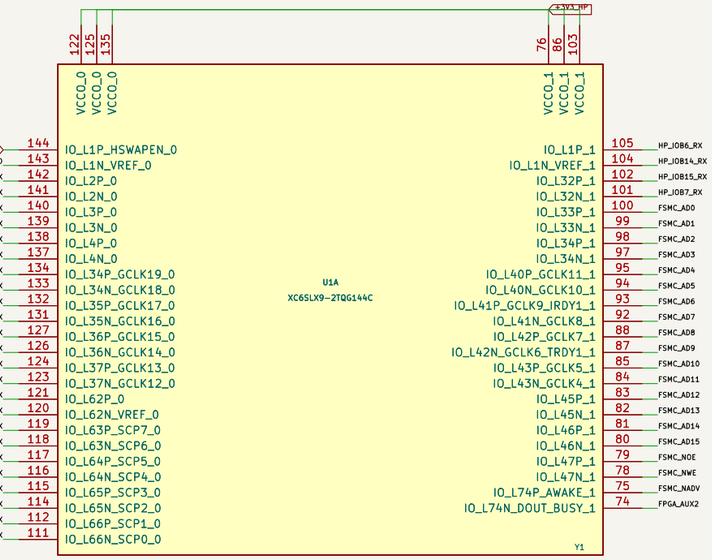
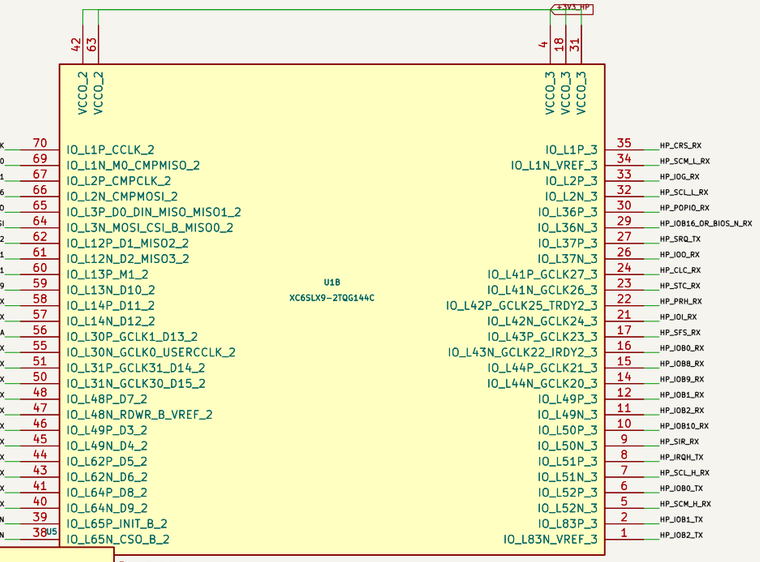
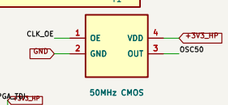
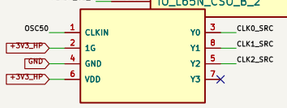
FPGA pin allocation / Master SPI / clocks / FPGA rails

Initial capture: review pin maps and draft circuits before PCB work.

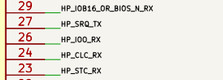


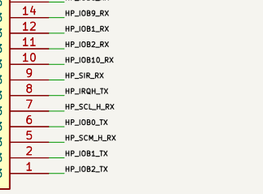


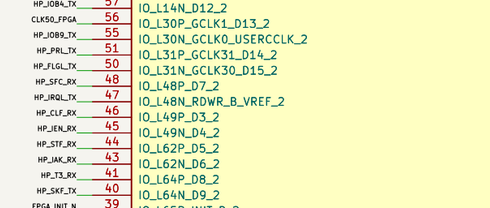


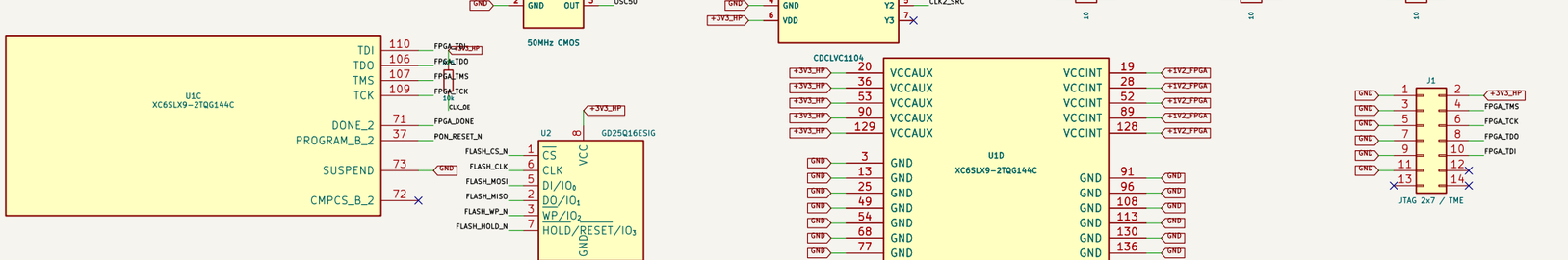


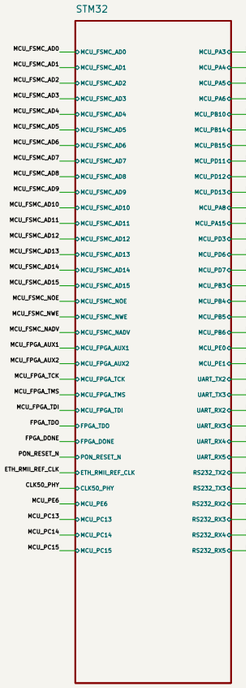


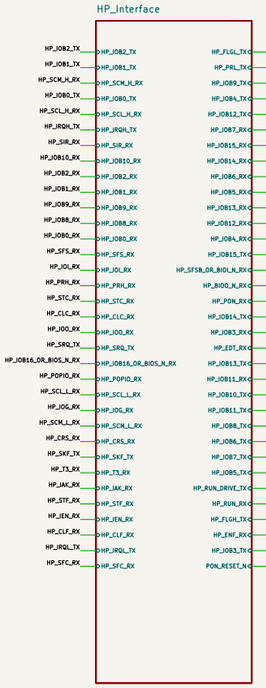


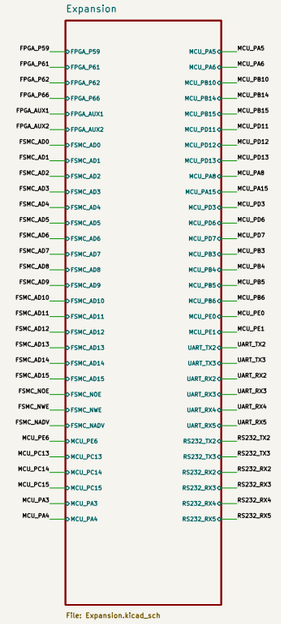












12.8mm component-height limit; THT from TME; HRO USB from JLCPCB
DRAFT – electrical and footprint review required

Sheet: /
File: HP21xx_IO.kicad_sch

Title: HP 21xx integrated I/O – FPGA

Size: A3	Date: 2026-11-08	Rev: 0.1 capture
KiCad E.D.A. 10.0.5		Id: 1/4

STM32F407VG / independent USB recovery domain

HRO USB-C / JLCPCB SMT assembly

U1001 STM32F407VGT6

J1001 HRO TYPE-C-31-M-12

U1002 GOODWORK USBLC6

J1003 SWD 2x10 / TME

J1004 BOOT0

J1005 DEV

U1005 SN75185DWR

J1002 SD-106M / CD10 WP11

Y1001 8MHz / CL12p

SW1001 RESET

+3V3_MCU / TLV62569 DBV (no PG)

H10 OPEN: pendant numbered pinout; current limit; /OE hardware gate; connector protection.

H06 OPEN: PHY line-side magnetics/straps/ESD and accepted MTJ-8B5X1 footprint need review.

Ethernet PHY/magnetics capture remains open; no fabricated cable pin numbers.

12.8mm component-height limit; THT from TME; HRO USB from JLCPCB
DRAFT — electrical and footprint review required

Sheet: /STM32/
File: STM32.kicad_sch

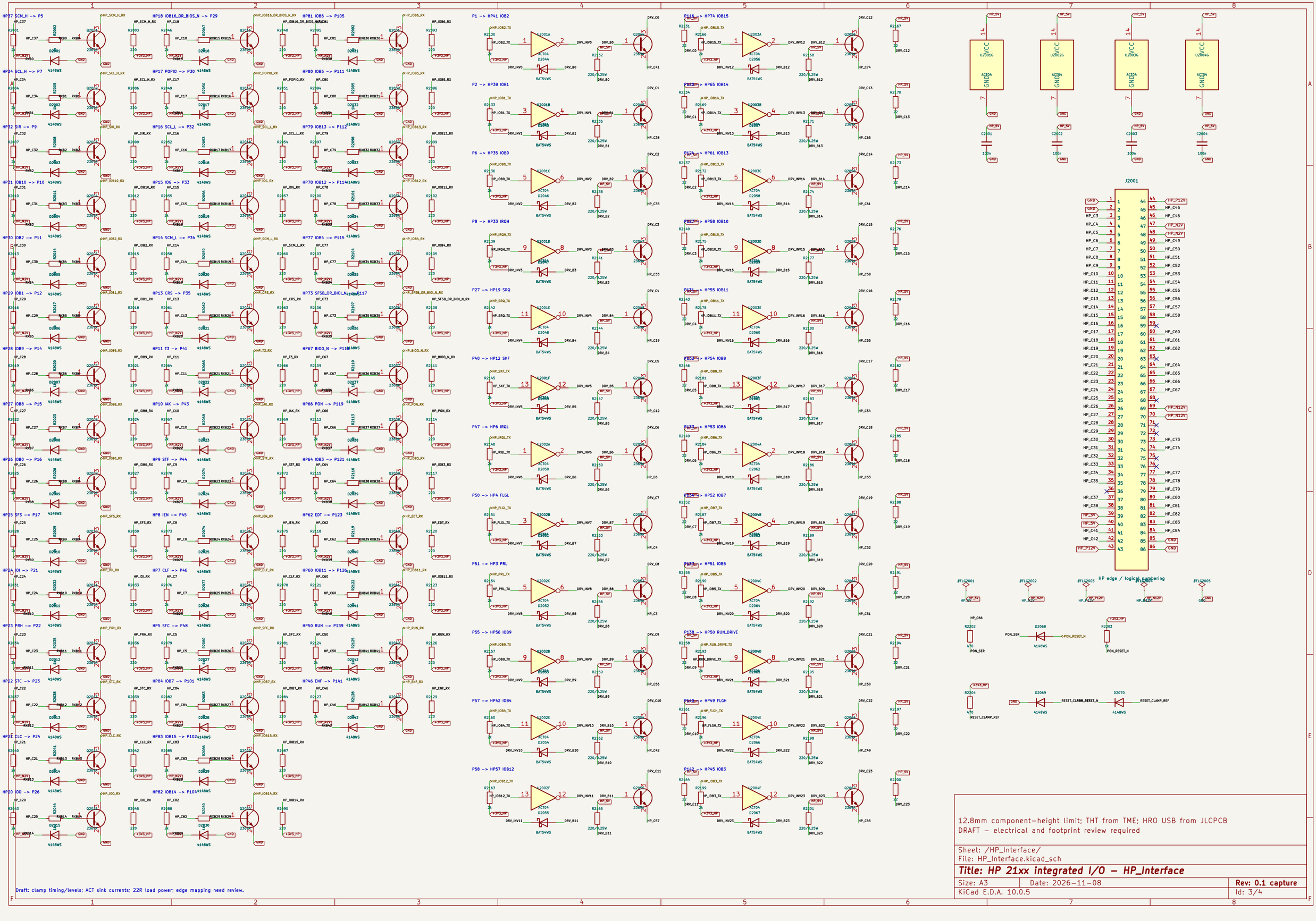
Title: HP 21xx integrated I/O — STM32

Size: A3	Date: 2026-11-08	Rev: 0.1 capture
KiCad E.D.A. 10.0.5		Id: 2/4

43 inverting receiver channels

24 ACT04 + emitter-output drivers

Driver supplies / rail entry / PON conditioner



Expansion / reusable FSMC / spare RS-232

Logical access pads only. HO2: final top connector, posts and numbered contacts remain open.

Do not drive shared FPGA/MCU nets until BOTH endpoints are released. RS-232 is bipolar.

FPGA_P59
FPGA_P61
FPGA_P62
FPGA_P66
FPGA_AUX1
FPGA_AUX2
FSMC_AD0
FSMC_AD1
FSMC_AD2
FSMC_AD3
FSMC_AD4
FSMC_AD5
FSMC_AD6
FSMC_AD7
FSMC_AD8
FSMC_AD9
FSMC_AD10
FSMC_AD11
FSMC_AD12
FSMC_AD13
FSMC_AD14
FSMC_AD15
FSMC_NOE
FSMC_NWE
FSMC_NADV

TPCD01 FPGA_P59
TPCD02 FPGA_P61
TPCD03 FPGA_P62
TPCD04 FPGA_P66
TPCD05 FPGA_AUX1
TPCD06 FPGA_AUX2
TPCD07 FSMC_AD0
TPCD08 FSMC_AD1
TPCD09 FSMC_AD2
TPCD10 FSMC_AD3
TPCD11 FSMC_AD4
TPCD12 FSMC_AD5
TPCD13 FSMC_AD6
TPCD14 FSMC_AD7
TPCD15 FSMC_AD8
TPCD16 FSMC_AD9
TPCD17 FSMC_AD10
TPCD18 FSMC_AD11
TPCD19 FSMC_AD12
TPCD20 FSMC_AD13
TPCD21 FSMC_AD14
TPCD22 FSMC_AD15
TPCD23 FSMC_NOE
TPCD24 FSMC_NWE
TPCD25 FSMC_NADV

MCU_PE6
MCU_PC13
MCU_PC14
MCU_PC15
MCU_PA3
MCU_PA4
MCU_PA5
MCU_PA6
MCU_PB10
MCU_PB14
MCU_PB15
MCU_PD11
MCU_PD12
MCU_PD13
MCU_PA8
MCU_PA15
MCU_PD3
MCU_PD6
MCU_PD7
MCU_PB3
MCU_PB4
MCU_PB5
MCU_PB6
MCU_PE0
MCU_PE1

TPCD26 MCU_PE6
TPCD27 MCU_PC13
TPCD28 MCU_PC14
TPCD29 MCU_PC15
TPCD30 MCU_PA3
TPCD31 MCU_PA4
TPCD32 MCU_PA5
TPCD33 MCU_PA6
TPCD34 MCU_PB10
TPCD35 MCU_PB14
TPCD36 MCU_PB15
TPCD37 MCU_PD11
TPCD38 MCU_PD12
TPCD39 MCU_PD13
TPCD40 MCU_PA8
TPCD41 MCU_PA15
TPCD42 MCU_PD3
TPCD43 MCU_PD6
TPCD44 MCU_PD7
TPCD45 MCU_PB3
TPCD46 MCU_PB4
TPCD47 MCU_PB5
TPCD48 MCU_PB6
TPCD49 MCU_PE0
TPCD50 MCU_PE1

UART_TX2
UART_TX3
UART_RX2
UART_RX3
UART_RX4
UART_RX5
RS232_TX2
RS232_TX3
RS232_RX2
RS232_RX3
RS232_RX4
RS232_RX5

TPCD51 UART_TX2
TPCD52 UART_TX3
TPCD53 UART_RX2
TPCD54 UART_RX3
TPCD55 UART_RX4
TPCD56 UART_RX5
TPCD57 RS232_TX2
TPCD58 RS232_TX3
TPCD59 RS232_RX2
TPCD60 RS232_RX3
TPCD61 RS232_RX4
TPCD62 RS232_RX5

TPCD63 +3V3_HP
TPCD64 +3V3_MCU
TPCD65 +1V2_FPGA
TPCD66 HP_5V
TPCD67 GND
TPCD68 HP_P12V
TPCD69 HP_N12V

This sheet does not invent a production connector footprint or contact allocation.

12.8mm component-height limit; THT from TME; HRO USB from JLCPCB
DRAFT — electrical and footprint review required

Sheet: /Expansion/
File: Expansion.kicad_sch

Title: HP 21xx integrated I/O — Expansion

Size: A3	Date: 2026-11-08	**Rev: 0.1 capture**
KiCad E.D.A. 10.0.5		Id: 4/4